

CS 451: Computer Graphics and Game Design

Course Code	CS 451
Course Name	Computer Graphics and Game Design
L-T-P-C	2-0-2-3
Prerequisites	Experience in writing computer-code.
Intended for	B.Tech. (all branches) II/III/IV year
Distribution	Elective for B.Tech. II/III/IV year
Approval	recommended by the 35th BoA (Aug 2020)

Course Contents

- **Introduction:** Motivation; Types of games, Different aspects of game design; Different components in a game; Game engines; Geometric primitives, 2D and 3D linear transforms, Homogeneous matrices; Examples of games. [5 Lectures]
- **Sprites and animation:** Different image formats; Polygon file formats; Creating sprites; Rigging; Animations using sprite-sheets; Animations using keyframes; Animation controllers. [6 Lectures]
- **Level design:** Scenes; Tiles, visual continuity in tiles; Adding objects to scene; Prefabs; Lighting, RGB space, transparency, texture mapping; Collectibles; Navigation and pathfinding. [6 Lectures]
- **World interaction:** Physics engines; Gravity simulation; Rigid body interaction; Collisions. [6 Lectures]
- **User interface:** Layout; Menu system; Visual components; Event system; Skins. [3 Lectures]
- **Audio:** Audio assets; Different audio formats; Audio mixing. [2 Lectures]

Laboratory Modules

Lab to be conducted on a 2-hour slot, in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course shall be appropriately sequenced for synchronization with the laboratory. The students will progressively design and write a 2D video game as part of the lab.

- Lab1-2: Installation of a game engine, e.g., Unity, familiarization of the GUI. Conceptualize the theme for a 2D game.
- Lab3-4: Character design, sprites, movement and character control.
- Lab5-7: Level design: design of the world in form of tiles along with interactive and collectible objects.
- Lab8-9: Design of interaction between the player and the world, optionally using the physics engine.
- Lab9-11: Design of menus and user interaction in mobile platform.
- Lab12: Insert audio.

Text Books

1. Nystrom Robert, **Game Programming Patterns**, 3rd Edition, Genever Benning, 2014

References

1. Paris Buttfield-Addison et al., **Unity Game Development Cookbook: Essentials for Every Game**, O'Reilly Media, 2019.